Kutty Banerjee

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Summary

- Specialize in GPUs, OS kernels, Memory Management, Thread Scheduling, Full stack tuning for Perf/Power, Profilers, Parallel Programming.
- Held various positions in GPU/CPU teams at NVIDIA. Was the SW Lead for Fermi GPU Architecture and Architect for Project Orion, NVIDIAs unified CPU/GPU effort.

Software Architect, NVIDIA [Dec 2010 ~ Present]

- **Project Denver – ARM CPU JIT**
  - Denver uses a **JIT** runtime to convert ARM instructions to native instructions. I maintain the **Branch Profiling, Translation Cache and Code Garbage Collection** code.
  - Implemented branch-to-assert and lazy assert-to-branch logic. Boosts SPECINT by upto 15%
  - Implemented Adaptive Branch Profiling framework to route hot translations to interpreter for a small time to regather branch weights and retranslate if weights shift.
  - Implemented **LockStat** GDB extension and an **Auto Deadlock** detection logic.
  - Implemented various stubs to **profile** transitions to and from interpreted / optimized code
  - Worked with various HW teams to design **Transparent Cluster Switching** in Denver microcode for Windows on Tegra.
  - Adding support for Generational compaction in **Code Garbage Collector**.
- **Android Stack Perf Optimizations for Denver CPU**
  - **Linux Kernel** : Implemented Large Page Suport for ARMv7 to automatically coalesce contig small pages into 64K pages. 5% IPC boost on SPECFP
  - **Linux Kernel** : Implemented **Thread Local Memory type** in kernel. TLB invalidates to this memory are not broadcast to other cores.
  - Patched **Kernel**, **Mozilla, Google V8 and Dalvik** cache maintenance layers to leverage Denver's cache layout and coherence model.
  - Patched **Dalvik** locking subsystem to utilize Denver's Transactional Memory system better.
  - Implemented **Memory Profiler** Extension to QEMU to study page access patterns for apps.
  - Investigating **V8 Browser Garbage Collection** changes to maximize Cache footprint and Prefetchability for Pointer chasing object code.
- **Project Orion (CUDA) – Cache Coherent CPU/GPU on a single SoC**
  - Designed and implemented proof of concept changes to AARCH64 Linux kernel and GPU kernel **MMU** layers to present a Unified Address Space between CPU and GPU for CUDA programs. Also added asynchronous GPU page fault handling.
- **Scalable MultiCore Research (Linux/ARM)**
  - **Khronos** : A **DataFlow OoO Processor** implemented in SW that takes in Serial Blocks of Code and **Parallelizes it and Schedules across separate Threads/Cores** after doing Depedency Analysis. Extracts Task/Data Parallelism, Parallel Reductions and I/O pipelining. Optimizes Scheduling to get best Data/Task colocality to minimize memory and cache traffic.
  - **Concurrency Utilities / Locks** : Variants of Biased Lock, Ticket Lock, SEQLOCK, Reader Writer Lock, Bulk Synchronous Parallelism utility. The focus is on reducing CAS/fence usage, minimizing cacheline traffic and improving lock fairness. **Tuned for ARMv7.**
  - **Task Parallel Library (C)** : Special focus on low latency low lock contention, CAS/fenceless intra thread communication. Wired up to aide Project Khronos above. **Tuned for ARMv7.**
  - **Task Parallel Library: (JAVA)** Special focus on **Dynamic Parallelism** and reducing lock contention on the requeue task paths. Also supports Pipelined Locks, Pipelined dependencies.
  - **GCC Attributes**: Patched 4.7.0 GCC to add an attribute to aid Static Lock Ordering Problem detection.

Senior System Software Engineer, NVIDIA [2005 June ~ Dec 2010]

- **GPU/Chipset Verification Lead**
  - The role was to reach out to the various HW unit leads, SW stake holders and make sure we had a coherent Chip Verification Strategy across HW/SW for future GPUs/Chipsets.
  - **PCIE Verification**: Drove changes to enable Directed Tests to run on Vista/XP for ASLM, ASPM, Standby, Hibernate, TDR, chipset deadlocks while reading CPU memory from GPU
  - **Power Mgmt Verification:** Designed a Probe Based Framework which allowed probes to be inserted into our Driver. Probes can introduce arbitrary waits/delays waiting for HW events. This allowed testing Power Transitions overlapped with interesting GPU states.
  - **Display Bandwidth Verification**: Coordinated with Display HW leads and SW leads to allow testing Display scan out while stressful GPU activities occur concurrently prior to tapeout in RTL and Functional Model. This allowed avoiding Bandwidth Starvation Bugs.
  - Increased coverage of trouble areas like **GPU context switching, Thread scheduling and ECC** using HW unit randoms and directed tests being fed real world instruction traces
  - Put in place Code Coverage System for Driver code using Bullseye.
- **Fermi GPU Driver CHIP Lead**
  - Drove the GPU kernel team from POR Architecture through RTL/Emulation/Bringup and Release. This included leading a 15 member team and interfacing with other SW and HW teams.
- **Fermi GPU Driver Power Management**
  - Implemented P-state switching support. Worked with HW to get BLCG/ELCG support. Designed Engine Level Power Gating code, team of 2 implemented the change.
  - Helped scope out work for a team of 3 to offload code to a microcontroller to avoid CPU interrupt latency problems.
  - Worked with HW and SW unit leads to draft the first proposal for Vista Workload Throttling. This was later postponed until the next GPU generation.
- **Fermi GPU Driver Thread Scheduling Lead**
  - Designed and implemented 85% of the Scheduler code in the driver. This unit was redesigned from scratch in Fermi to avoid latency problems of the previous generation.
  - Implemented a Pipelined Scheduling Mode that allowed queued tasks to prefetch, TLB and cache warmup before the threads are executed. **Microsoft Vista DX10** feature.
  - Implemented separate **scheduling policies** to **saturate PCIE b/w for Graphics threads** and minimize Launch Latency for CUDA/PhysX/Video threads.
  - Designed the Error Recovery Subsysten and Power Aware Scheduling Subsystem. Mentored team members to implement the actual code (**Vista TDR** subsystem)
  - Implemented **Mid Buffer Preemption of Graphics work on Vista** for better User Experience
  - Debugged Perf and Latency bottlenecks
- **Fermi GPU Driver Kernel MMU / Memory Management**
  - Helped the MMU lead wire up memory management layer of the Fermi GPU kernel. This includes basic support for Multi level Page Table setup code, TLB invalidation logic, TLB invalidation batching optimizations, NOMMU mode support, fixed bugs.
- **Driver Non Stall Interrupts**
  - Implemented low latency lockless interrupt code path and ported all GPU notifiers to use this. This interrupt line is very rarely disabled and has low dependency on non ISR code.
- **Curie Architecture GPU Kernel Driver**
  - Debugged all parts of the GPU kernel, fixed bugs and triaged perf bottlenecks.

Education

MS Computer Science, Worcester Polytechnic Institute, MA [2003 Aug ~ 2005 May]

- **Thesis**: Remote Execution for 3D Graphics. A distributed 3D Rendering pipeline (RMESA) to offload floating point intensive operations to a cluster to save Power on Mobile Devices
- Recipient of **WPI Institute Fellowship Award** for outstanding research work.

BS Computer Science, University of Mumbai [1999 July ~ 2003 May]

Publications

- **PowerSpy: Fine-Grained Software Energy Profiling for Mobile Devices**
  Kutty Banerjee, Emmanuel Agu
  International Conference on Wireless Networks, Communications and Mobile Computing, 2005
- **A Middleware Architecture for Mobile 3D Graphics**
  Emmanuel Agu, Kutty Banerjee, Shirish Nilekar, Oleg Rekutin, Diane Kramer
  International Journal of Parallel, Emergent and Distributed Systems Volume 21, Issue 3, 2006
- **Estimating Mobile Memory Requirements and Rendering Time for Remote Execution of the Graphics Pipeline**
  Kutty Banerjee, Fan Wu, Emmanuel Agu - Proceedings of Eurographics 2005
- **Remote Execution for 3D Graphics on Mobile Devices**
  Kutty Banerjee, Emmanuel Agu
  International Conference on Wireless Networks, Communications and Mobile Computing, 2005
- **A Middleware Architecture for Mobile 3D Graphics**
  Emmanuel Agu, Kutty Banerjee, Shirish Nilekar, Oleg Rekutin, Diane Kramer
  Third International Workshop on Mobile Distributed Computing, 2006

US Permanent Resident